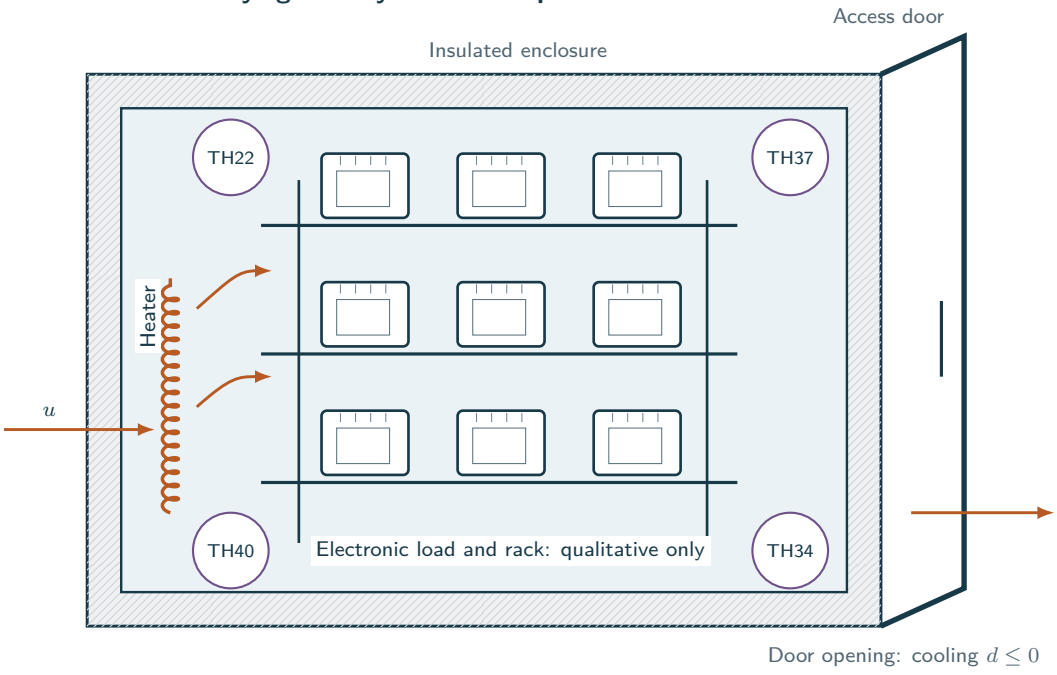
Burn-in chamber: physical picture and reduced model

01 / Chamber, four measurement channels and the meaning of the FOPDT approximation

Illustrative cutaway: geometry and sensor positions are not recovered from the thesis

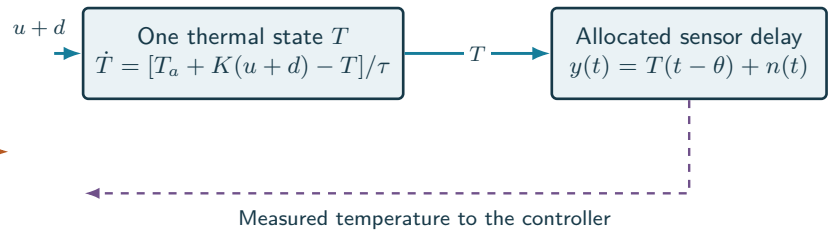


What the identified source model represents

Combined heating / measurement response:

$$G(s)H(s) = \frac{\bar{K}\, e^{-\theta s}}{\bar{\tau} s + 1}.$$

Mean-temperature parameters: $\bar{K} = 7.624$, $\bar{\tau} = 151.6$ s, $\theta = 8.78$ s. The effective command is not calibrated in watts.



A separate four-channel synthetic dataset

TH22, TH37, TH40 and TH34 are named channels; their mean and spread can be inspected. The one-state feedback simulation does not resolve a spatial temperature field. Mean-temperature tracking alone cannot certify uniformity.

Physical interpretation boundary

The cabinet, rack, heating element and sensor locations explain thermal and measurement paths; they are not recovered hardware dimensions, wiring or a safety design. No actual equipment layout or measured temperatures are supplied by this illustration.

Declared simulation choices

Ambient $T_a = 30\,^{\circ}\mathrm{C}$; reference $65\,^{\circ}\mathrm{C}$; effective command $0 \leq u \leq 10$; baseline sample period 1 s. Delay allocation to the sensor, door-loss amplitude and four-channel field generation are assumptions, separate from source-reported FOPDT parameters.

Blue arrows: model/signal propagation. Orange arrows: qualitative heating/cooling paths. Purple sensors: named measurement channels; their placement is illustrative. See source_traceability.md and models.py.

Original explanatory drawing for ThermoFuzzy-PID. Source traceability: Cordeiro (UFC, 2021), documented repository equations and explicit reconstruction choices.

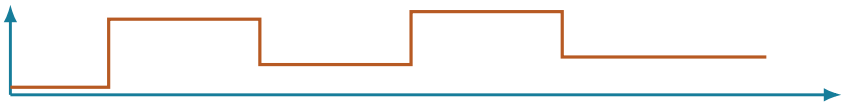
Identification: excite, estimate, then challenge the model

02 / Source-reported parameter sets and separate synthetic identification experiments

Source parameter register: four distinct FOPDT models

Case	K	τ [s]	θ [s]
TH37	8.7135	326.18	13.677
TH34	7.7473	177.95	13.064
Mean measured temperature	7.624	151.6	8.78
Mean estimated parameters	8.028	186.5	12.5

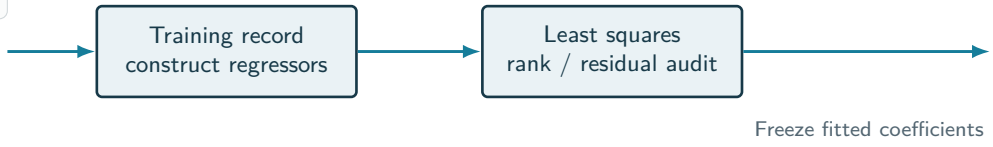
The mean of sensor responses and the mean of fitted parameters need not have the same dynamics. TH22 and TH40 are not independently identified here.



Illustrative excitation u_k: record sample times, ambient offset, input units and delay convention

Two explicitly different estimators

Notebook 01 fits an analytical FOPDT step response to synthetic data. Notebook 05 uses the restricted known-integer-delay ARX model below. Neither experiment recreates the historical MATLAB fit or recovers missing original input/temperature records.



ARX(1,1) with a known integer delay m

$z_k = y_k - T_a, \quad z_k = a z_{k-1} + b u_{k-1-m} + \varepsilon_k.$

$\Phi_k = [z_{k-1} \quad u_{k-1-m}], \quad \hat{\beta} = \arg\min_{(a,b)} \|\Phi\beta - z\|_2^2.$

$\hat{\tau} = -\Delta t / \ln \hat{a}, \quad \hat{K} = \hat{b}/(1-\hat{a}), \quad \hat{\theta} = m\Delta t.$

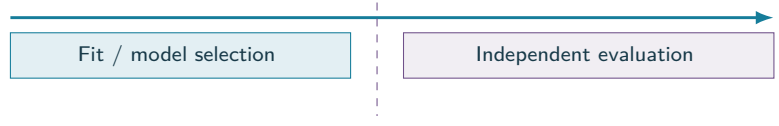
Mapping requires $0 < \hat{a} < 1$, $\hat{b} > 0$, full-rank excitation and compatible units. The integer-delay formulation does not reproduce a fractional delay exactly.

Recommended extension: independent evaluation

One step: predict with the measured previous output. A low residual can hide dynamic mismatch.

Free run: feed earlier predictions back into the model. Check accumulated bias, delayed onset and long-term gain.

Not executed by notebooks 01/05: their fitted residual RMSE is in-sample. Freeze model choices before an independent test.



Recommended split; not executed by notebooks 01/05

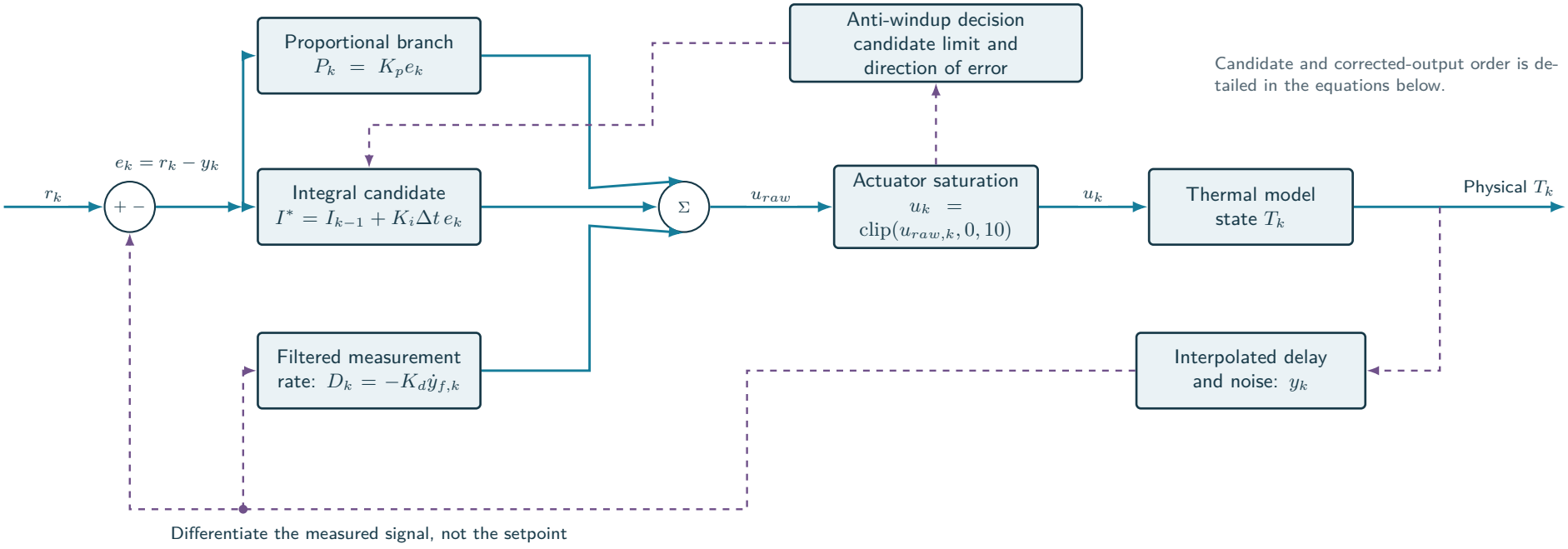
Unresolved historical inputs

Original synchronized heater drive, raw sensor readings and sufficiently documented sampling/validation protocol are not supplied. The diagrams and synthetic fits do not establish the thesis's reported fit percentages.

The command trace is a conceptual excitation sketch, not a dataset. Parameter values follow the repository's source register; estimator equations follow identification.py and the notebook-specific protocol. Original explanatory drawing for ThermoFuzzy-PID. Source traceability: Cordeiro (UFC, 2021), documented repository equations and explicit reconstruction choices.

PID implementation: gain convention, filtering and saturation

Independent gains: heater/°C, heater/(°C s), heater s/°C. Derivative filter: $\dot{y}_{f,k} = a\dot{y}_{f,k-1} + (1-a)(y_k - y_{k-1})/\Delta t$, with $a = \tau_f/(\tau_f + \Delta t)$.



IMC source-derived tuning

$$K_p = \frac{\tau + \theta/2}{K(\lambda + \theta)}; \quad T_i = \tau + \theta/2.$$

$$T_d = \frac{\tau\theta}{2\tau + \theta}.$$

Parallel gains: $K_i = K_p/T_i$, $K_d = K_p T_d$. Time constants and gains must not be interchanged.

Conditional integration

Compute $u^* = P_k + I^* + D_k$. Retain I_{k-1} if $u^* > u_{max}$ with $e_k > 0$, or $u^* < u_{min}$ with $e_k < 0$. Otherwise accept I^*. Then recompute and clip the command.

Back-calculation option

$$I_k = I^* + \frac{\Delta t}{T_t}\left[\mathrm{clip}(u^*) - u^*\right].$$

Recompute $P_k + I_k + D_k$, then clip. The implementation requires $T_t \geq \Delta t$. Its numerical choice is an assumption.

Control semantics checked against FixedPID._calculate and imc_pid_gains. Fresh controller objects are required for independent runs because integral, derivative and measurement histories persist. No hardware protection is modeled.

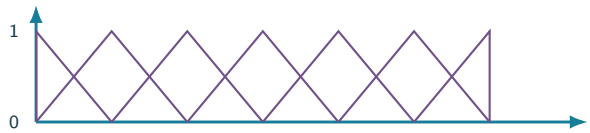
Original explanatory drawing for ThermoFuzzy-PID. Source traceability: Cordeiro (UFC, 2021), documented repository equations and explicit reconstruction choices.

Mamdani gain scheduling: memberships, omissions and memory

04 / Exact sparse consequents, declared membership reconstruction and three independent outputs

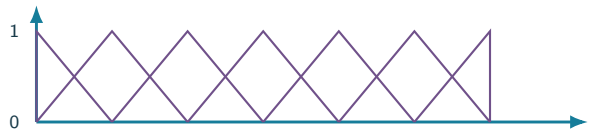
$\Delta e_k = e_k - e_{k-1}$; both inputs are clipped to their domains before fuzzification.

Error magnitude



$|e| \in [0, 40]\,^{\circ}\mathrm{C}$

Per-sample error change



$\Delta e \in [-40, 40]\,^{\circ}\mathrm{C}$

From activation to a normalized gain
Minimum antecedent strength; maximum consequent aggregation. Clip output triangles; average all maximizers on $[0, 1]$ (201 points by default): mean of maxima, not centroid.

Proportional output K'_p

	NB	NM	NS	Z	PS	PM	PB
NB	M	–	MS	M	MS	–	B
NM	MS	–	MS	M	MS	–	B
NS	S	–	MS	M	MS	–	B
Z	S	–	MS	M	MS	–	B
PS	S	–	MS	M	MS	–	B
PM	MS	–	MS	M	MS	–	B
PB	M	–	MS	M	MS	–	B

Columns: $|e|$ category; rows: signed Δe category

Integral output K'_i

	NB	NM	NS	Z	PS	PM	PB
NB	B	–	S	S	–	–	S
NM	MB	–	S	S	S	–	S
NS	M	–	S	S	S	–	S
Z	S	–	S	S	S	–	S
PS	S	–	S	S	S	–	S
PM	MB	–	S	S	S	–	S
PB	B	–	S	S	–	–	S

Columns: $|e|$ category; rows: signed Δe category

Derivative output K'_d

	NB	NM	NS	Z	PS	PM	PB
NB	S	–	MS	M	MB	–	B
NM	S	–	MS	MS	M	–	M
NS	S	–	S	S	MS	–	S
Z	S	–	S	S	S	–	S
PS	S	–	S	S	MS	–	S
PM	S	–	MS	MS	M	–	M
PB	S	–	MS	M	MB	–	B

Columns: $|e|$ category; rows: signed Δe category

Omitted does not mean zero
Gray NM/PM columns are unspecified in the source table. "–" marks an omitted gain. If a gain has no active consequent, retain its previous normalized value (initially 0.5). Each gain has separate memory.

Map each normalized output

$K_j = K_{j,min} + (K_{j,max} - K_{j,min})K'_j.$

$K_{min} = (.35, .0008, 0);$
$K_{max} = (.90, .045, 2).$

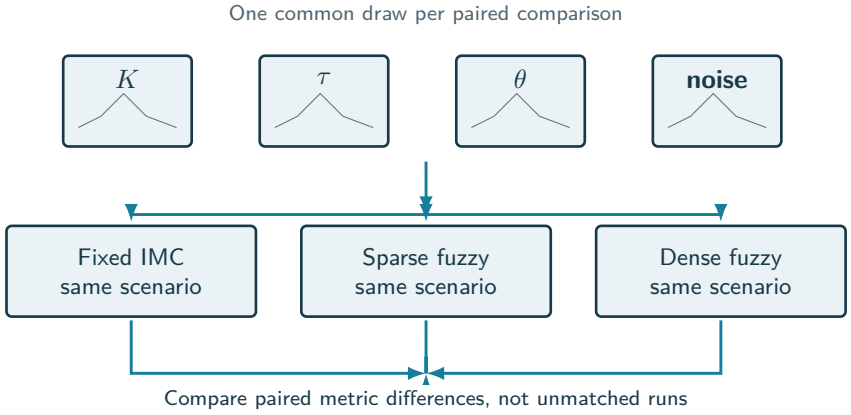
The output vector orders the parallel proportional, integral and derivative gains.

Reconstruction versus extension
Equal membership spacing is assumed. NB–PB label categories even on the nonnegative magnitude axis. Dense rules interpolate omissions; smoothing and IMC blending are separate research extensions. None is a recovered hardware setting.

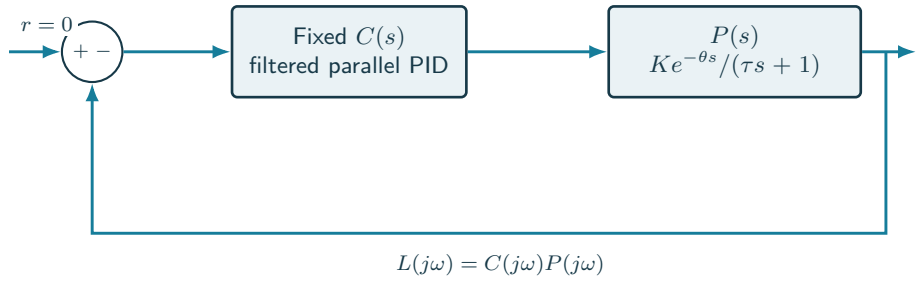
The three matrices are generated directly from EXACT_SPARSE_RULES in fuzzy.py and retain the thesis's omissions. Membership sketches show the declared triangular geometry; the existing computed surface figures remain separate evidence.
Original explanatory drawing for ThermoFuzzy-PID. Source traceability: Cordeiro (UFC, 2021), documented repository equations and explicit reconstruction choices.

Robustness: paired uncertainty and a separate linear loop audit

A. Nonlinear / sampled comparison
Freeze a controller design, draw synthetic plant parameters, then run every controller against the same parameter draw, noise seed, limits and disturbance schedule. Pairing reduces comparison noise; it does not validate the chosen probability distributions.



B. Fixed, unsaturated return-ratio model
Analyze one fixed PID and one FOPDT plant, with the reference held at zero for the loop test. Report the frequency grid and crossover method. Scheduling and clipping are omitted; margins do not certify the nonlinear fuzzy loop.



Report a vector of consequences
Rise time and overshoot before disturbances; full-horizon IAE; command total variation; saturation fraction; post-closure recovery with a sustained tolerance band. Lower error can require more actuator movement. Record both.

Sensitivity and margins
$S(j\omega) = 1/[1 + L(j\omega)], \quad M_s = \max_{\omega \in \Omega} |S(j\omega)|.$

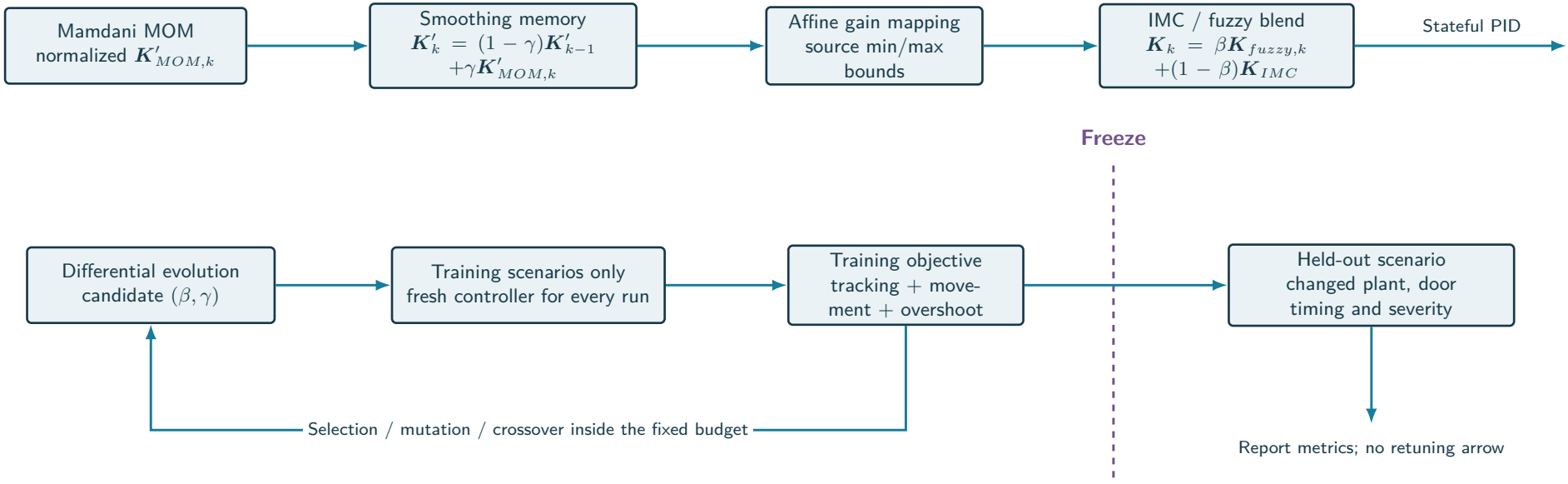
Gain and phase margins use the first detected crossings on an explicit grid. No crossing on that grid is not proof of an infinite physical margin. Inspect additional crossings and Nyquist behavior when needed.

Missing physical evidence
A single thermal state does not predict sensor-to-sensor uniformity, component temperature, heater calibration or equipment interlocks. Synthetic parameter sweeps and linear-loop margins address declared numerical models only.

The distributions drawn above are conceptual symbols, not fitted chamber distributions. Numerical uncertainty and frequency-response plots remain generated by the notebooks and robustness.py, with their original seeds and model assumptions.
Original explanatory drawing for ThermoFuzzy-PID. Source traceability: Cordeiro (UFC, 2021), documented repository equations and explicit reconstruction choices.

Optimization: tune on training scenarios, freeze, then evaluate

Design variables: $\beta \in [0, 1]$ and $\gamma \in [0.05, 1]$ for the optimization notebook. $\beta = 1$ uses fuzzy gains only; $\beta = 0$ uses the fixed IMC gains. Smaller γ slows gain changes. These mechanisms are post-thesis extensions.



What is held constant
Training scenarios, 3 s simulation sample period, actuator limits, random seeds and evaluation budget. Start each trajectory with a fresh controller; otherwise integral and scheduling history contaminate comparisons.

What the optimum means
Best sampled training objective under the declared search, not a global optimum or a physical performance guarantee. Preserve the evaluation count and compare alternatives under matched information and computation.

What must be reported
Independent IAE, overshoot, recovery and total variation, plus adverse outcomes. The historical 22% / 3% observations are reported source results; they are not reproduced by this synthetic search.

Workflow follows notebook 04 and FuzzyPID.step. The evaluation scenario is separated from candidate selection. Search settings, parameter distributions and the synthesized dense rule table remain documented research choices.
Original explanatory drawing for ThermoFuzzy-PID. Source traceability: Cordeiro (UFC, 2021), documented repository equations and explicit reconstruction choices.